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Accelerating resident research within quantitative collaboration units in academic healthcare

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With increased access to biomedical and electronic health records data and the complexity of research questions, individuals in residency programmes who aim to conduct research require specialized educational programmes and biostatistics support. Biostatistics collaboration units in academic health centres often work with residents to conduct data-intensive research. These units face numerous challenges related to providing training in statistical literacy and collaborating on resident-led research within very restricted timelines. Since 2019, the Duke Biostatistics, Epidemiology, and Research Design (BERD) Methods Core has supported over 247 resident-led projects by developing tools and resources to address these challenges. This manuscript presents novel processes and training materials that other institutions can use to help biostatistics collaboration units effectively support resident training programmes. We provide a framework to support the development of collaborative teams, along with specialized training materials for residents who collaborate with these teams.

KEYWORDS
biostatistics collaboration, collaboration resources, collaborative process, resident training, resident-led research, team science

1 | INTRODUCTION

Building collaborative research teams and incorporating research training into residency programmes within academic health centres are essential for enhancing career opportunities and fostering innovation within evolving healthcare landscapes (Anderson, 2006; Hurst et al., 2019; Institute of Medicine, 2004). Engaging in research improves satisfaction with the residency experience and aids in developing future leaders in academic medicine (Takahashi et al., 2009). Individuals in residency programmes engaged in research as part of their training can benefit from collaborative research experiences and tailored learning to advance their career goals (Khalil et al., 2022; Permar et al., 2020). Academic health centres must provide the necessary tools and support to facilitate this (Blanchard et al., 2018; Price Rapoza et al., 2022; Welty et al., 2013).

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Medical professionals need a fundamental understanding of research design and analytics, which is crucial for evidence-based medicine (Arias et al., 2017; Oster & Enders, 2018; Sackett et al., 1996). Key competencies include identifying the strengths and limitations of study designs, recognizing when to involve a quantitative expert, and appreciating reproducible research (Enders, 2011; Enders et al., 2017; Oster et al., 2015; Oster & Enders, 2018). However, many residents lack the statistical knowledge and confidence to interpret study results (Araoye et al., 2020; Best & Laskin, 2013; Bookstaver et al., 2012; Couture et al., 2020; Susarla & Redett, 2014; Windish et al., 2007). Biostatistics collaboration units that typically partner biostatisticians with investigators conducting research must account for these gaps in knowledge when developing processes to effectively collaborate with residents.

With increased access to biomedical and electronic health records data and the complexity of research questions, residents must have access to biostatistics collaboration and statistical literacy training. The biostatistician in the collaboration provides expertise in choosing the study design, sample size, and endpoints that can answer these complex research questions, and this guidance provides training for the rest of the team through example. Without biostatistics collaboration and statistical literacy training, projects may lack sufficiently developed research questions, rigour, reproducibility and statistical analyses (Bland, 2004; Brown et al., 2018; Gunel Karadeniz et al., 2019). Despite several proposed curricula for statistical education (Arias et al., 2017; McLaughlin & Kang, 2017; Windish, 2011), a gap persists in covering fundamental competencies, particularly statistical collaboration and reproducible research (Oster et al., 2020). In addition, collaboration units must work under the time constraints of resident trainees, who often need to assimilate their learning to develop data-intensive research questions rapidly.

This manuscript provides the foundational framework to address the problem of developing biostatistics collaboration teams dedicated to supporting resident training programmes. This work aims to support biostatistics collaboration units in overcoming two main challenges with resident-led research: (1) building a biostatistics team to mentor, train, and collaborate with residents on projects under time constraints and (2) training residents in clinical research to facilitate collaboration with the biostatistics team. We do this by reviewing the processes and training material developed by the Duke Biostatistics, Epidemiology, and Research Design (BERD) Methods Core, which has supported over 247 resident-led research projects. First, we describe the collaborative team model the BERD Core uses to support resident-led research. Next, we provide a roadmap for the collaboration process and template documents that support collaboration between residents and biostatisticians. Finally, we present flexible training modules that can be used to help residents better engage with biostatisticians on data-intensive research projects. We aim to provide biostatistics collaboration units with tools and training materials to facilitate engagement with resident-led research and to train residents in clinical research and collaboration.

2 | BIOSTATISTICS COLLABORATION TEAM TO SUPPORT FOR RESIDENTS

The BERD Core comprises over 40 staff and faculty biostatisticians and data scientists who partner with more than 20 research groups in the Duke School of Medicine and Duke University Health System (Desai et al., 2022). These partnerships are with Duke clinical research units such as the Global Health Institute, the Center for AIDS Research and the Department of Pharmacy, Neurosurgery, Pediatrics, among others. Biostatisticians in the BERD Core have partnered with residents across the School of Medicine on more than 247 data-driven research projects, resulting in over 77 publications since 2019. Funding for a biostatistician's effort may come from external grants, but often for resident-led research, biostatistics effort is supported by internal departmental funding.

We use the BERD Core's longest standing collaboration dedicated to supporting resident-led research as a case illustration of a successful biostatistics collaboration team. The collaboration with Duke University Hospital's Department of Pharmacy began in 2012 and has supported more than 100 resident-led projects since 2017. The Department of Pharmacy's residency programme consists of a generalized first-year programme, followed by an optional second year programme focusing on a pharmacy-specific specialty practice area. Residents in both years are required to complete a research project starting in July with the goal of dissemination by the following June. Consequently, these projects often need to occur simultaneously, posing a significant challenge for partnering biostatisticians. We have found that managing this workload is most effective when distributed across multiple biostatisticians.

In 2022, the Department of Pharmacy requested collaboration for 18 residents. The BERD Core found it optimal to divide the work among five staff biostatisticians, each contributing 15%–25% effort with varying levels of expertise and experience, one faculty member at 30%–40% effort, as well as two student interns who worked flexibly up to 20 h a week based on project needs. This allocation was determined by annual assessment of the residency programme's requirements and iteratively optimizing effort allocations to align with the group's needs. Moreover, at the start of the residency year, the residency programme leaders coordinate with the biostatistics team to assess the timelines and complexity of all projects so that projects can be effectively assigned to biostatisticians. The goal is to allocate effort to ensure that each biostatistician has a diverse range of projects with varying complexity and deadlines. It also enhances efficiency to assign projects with similar methods or clinical domains to the same biostatistician.

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3 | A ROADMAP FOR THE COLLABORATION PROCESS

Once appropriate biostatistics effort is in place, it is important to establish a clear and transparent timeline for residents and biostatisticians to follow. Appendix S1 presents the 'roadmap' that was developed to support a 12-month residency programme. This roadmap includes 11 steps that residents and biostatisticians take in order to reach the project goals within the established timeframe. The first step is for residents to meet with the biostatisticians on the team to discuss the study design for protocol development. In Step 2, biostatisticians provide documentation and a workshop on data collection using reproducible databases (e.g., Harris et al., 2009, 2019), which facilitates the third step, created to ensure residents know how to structure data collection in order to optimize collaboration with a biostatistician. The roadmap then specifies when data collection should begin and end, provides guidance for developing a statistical analysis plan and ends with a reminder to include biostatisticians as co-authors. Monthly meetings between the biostatistics team and residency programme leadership helps to ensure residents adhere to project timelines and resolve any issues promptly.

To help residents navigate the roadmap presented in Appendix S1, four templates were created for protocol development, data collection, statistical analysis plans and statistical reports. Appendix S2 provides a template that guides residents through submitting a protocol to the Duke Institutional Review Board. The study protocol includes data dictionaries which outline how measures will be collected and from which data sources. Appendix S3 provides a template data dictionary that is given to the residents who then use it to develop their database in Step 3 of the roadmap. Biostatisticians provide hands on support to residents as they complete these templates.

Once the residents have identified and described the data they intend to collect, finalized their protocols, and built their databases, biostatisticians will develop an agreed-upon statistical analysis plan (SAP) aligned with the research project's goals. Since biostatisticians are typically the primary author of SAPs, a template for this document was initially developed to train junior biostatisticians (Pomann et al., 2020). Appendix S4 presents a revised template designed specifically to help pharmacy residents better understand the SAPs that biostatisticians develop. In addition to a SAP template, Appendix S5 provides an analytic report template that is shared with residents to provide them with a better understanding of how the results will be presented when the biostatistician completes the analyses.

4 | JUST IN TIME TRAINING MATERIALS FOR RESIDENTS

In addition to the roadmap and templates presented, the BERD Core provides residents with optional just-in-time training materials that are publicly available (https://bit.ly/introtores). The BERD Core designed these training materials to fill curricular gaps in residency education. Notably, the educational content emphasizes the value of reproducible research and biostatistics collaboration (Oster et al., 2020). The content also encourages engagement by being problem-based, relevant, and applicable to clinical decision-making.

The BERD Core collaborated with instructional designers to align the design of the materials with best practices for digital learning today. The online modules are intentionally brief to minimize interference with residents' demanding clinical training schedules. Also, the material allows for easy pairing of online content with customizable in-person workshops tailored specifically to the clinical discipline of the audience. Lastly, the materials are well-organized, succinct, readily available, and can be accessed in a just-in-time manner as needed. The following subsections describe the content and results from a study to evaluate the materials.

4.1 | Training curriculum content and format

The four publicly available online modules introduce fundamental concepts of clinical research. Table 1 provides detailed information about what each module covers on four topics: (1) Introduction to Research and Design, (2) Formulating the Research Question, (3) Null and Alternative Hypotheses and (4) Study Design and Data Collection. Each module ranges from 10 to 20 min in length, with the total time commitment for all four being approximately 1 h. This online content was designed to teach fundamental competencies in an asynchronous, standalone format, enabling learners to access it in a self-directed manner as needed. Residents can watch these modules at any time before, during or after their research projects begin. The content can also be easily paired with in-person workshops that go into more depth about specific case studies that residents are currently working on or are interested in learning about.

4.2 | Training material evaluation

A small study was conducted in 2021 to evaluate the online modules among residents, trainees and programme leaders in the Department of Neurosurgery and the Division of Plastic Surgery at Duke University. The Institutional Review Board deemed this study exempt, as it focused on the efficacy of educational or instructional strategies. We shared the link to the curriculum with leaders from the Integrated Plastic and

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TABLE 1 Fundamental statistical competencies that are covered in the online training modules.

Module	Topics covered	Competencies addressed
Introduction to research and design	• Life cycle of clinical research • Research integrity, transparency and reproducibility	• Recognize limitations in statistical competency and realize when it would be best to involve a professional statistician • Understand the reasons for performing research that is reproducible from data collection through publication of results
Formulating the research question	• Research question development • Considerations and limitations in data driven research questions	• Recognize limitations in statistical competency and realize when it would be best to involve a professional statistician
Null and alternative hypotheses	• Definition and importance of the primary outcome • Null and alternative components of a statistical hypothesis • Types of errors	• Recognize limitations in statistical competency and realize when it would be best to involve a professional statistician • Understand the basic principles and practical importance of hypothesis testing and type I and type II errors
Study design and data collection	• Observational and experimental studies • Best practices for collecting data • Sample size considerations	• Recognize limitations in statistical competency and realize when it would be best to involve a professional statistician • Identify the strengths and limitations of study designs for addressing a clinical or translational research question • Understand the value of data quality and data management • Understand the components of sample size, power and precision

Reconstructive Surgery Residency Program and the Neurosurgery Residency Program. These leaders disseminated the curriculum to 27 residents and select medical students in their department and encouraged them to review the materials, provide feedback via an online survey and attend an hour long in-person workshop with a biostatistician. The workshop illustrated the content of the online modules using two or three resident research projects as case examples. The feedback survey was designed to assess the utility, quality and satisfaction of the online modules. Participants were asked about the number of modules watched, and if all four were not viewed, the reason for not doing so. Participants were also asked to rate the modules' length, the information's relevancy and satisfaction with the content.

Twenty participants responded to the survey. Of these, 14 (70%) were residents, and six (30%) were non-residents. The response rate among residents was 52% (14 of 27); however, we could not assess the overall survey response rate because the clinical leaders shared the curriculum and survey with select medical students and this activity was not tracked. Sixteen respondents (80%) reported having more than a little amount of prior experience conducting clinical research. Eighteen participants (90%) watched all four modules, while one (5%) watched three modules, and one (5%) watched no modules. The only reported reason for not watching all four modules was lack of time. Among the 18 participants who provided feedback, most (89%) said the modules were just the right length, all 18 (100%) found the content somewhat or very relevant and 14 (78%) were somewhat or extremely satisfied with the content (Table 2).

5 | DISCUSSION

Biostatistics collaboration units in academic healthcare settings need to support research training when partnering with residents. Residents can influence scientific progress and healthcare improvement by conducting data-intensive research in partnership with biostatisticians. Cultivating critical evaluation skills for published studies and establishing effective collaborations with quantitative experts is crucial for advancing evidence-based medicine and public health. This manuscript presents resources and training materials that collaboration units can use to help residents gain these skills and conduct meaningful research.

A primary challenge in supporting resident-led research within a biostatistics collaboration unit is establishing the right team with sufficient funding and flexible effort allocations to meet the requirements of the residency programme. While it is ideal for residents to be partnered with clinical or scientific mentors possessing research funding to address the quantitative collaboration needs, the number of residents aiming to do research will undoubtedly outnumber available funded projects for them to lead. Even if funding is in place, it can be challenging for biostatisticians to allocate effort effectively. This challenge is mitigated by having multiple biostatisticians who can work together to efficiently split the effort.

Another major challenge is the limited amount of time residents have for research activities. The BERD Core has addressed this challenge by developing a clear roadmap of the collaborative process with specific timelines for project milestones and regular check-ins with leadership to ensure that projects remain on target. In addition, we provide the residents with just-in-time online training modules that facilitate the collaborative process by introducing them to fundamental principles and limitations in clinical research and statistical competencies at a high level, rather

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TABLE 2 Summary of survey responses providing feedback on the online modules.

	Overall ($n = 20$)	Non-resident ($n = 6$)	Resident ($n = 14$)
Survey duration (minutes), median [IQR]	4.2 [2.6–4.8]	4.6 [4.4–5.1]	3.6 [2.5–4.7]
Age (years), median [IQR]	29.5 [27.8–32.2]	30 [25–47]	29.5 [28.2–31]
Race, *n* (%)			
American Indian, Alaska Native, First Nations, or Indigenous	1 (5)	1 (16.7)	0 (0)
Asian or Asian American	2 (10)	2 (33.3)	0 (0)
Black or African American	3 (15)	1 (16.7)	2 (14.3)
Hispanic, Latino/a/x, or Spanish origin	1 (5)	0 (0)	1 (7.1)
White	13 (65)	2 (33.3)	11 (78.6)
Gender, *n* (%)			
Man	9 (45)	1 (16.7)	8 (57.1)
Woman	11 (55)	5 (83.3)	6 (42.9)
Residency year, *n* (%)			
First year	2 (14.3)	–	2 (14.3)
Second year	3 (21.4)	–	3 (21.4)
Third year or beyond	9 (64.3)	–	9 (64.3)
How much prior experience do you have conducting clinical research?, *n* (%)			
A great deal	1 (5)	1 (16.7)	0 (0)
A lot	4 (20)	2 (33.3)	2 (14.3)
A moderate amount	11 (55)	1 (16.7)	10 (71.4)
A little	4 (20)	2 (33.3)	2 (14.3)
Number of videos watched, *n* (%)			
None	1 (5)	0 (0)	1 (7.1)
3	1 (5)	0 (0)	1 (7.1)
4	18 (90)	6 (100)	12 (85.7)
How would you rate the length of the videos?, *n* (%)			
Just the right length	16 (88.9)	5 (83.3)	11 (91.7)
Too long	2 (11.1)	1 (16.7)	1 (8.3)
Missing	2 (.)	0 (.)	2 (.)
How easy was it to follow the information in the videos?, *n* (%)			
Extremely easy	12 (66.7)	5 (83.3)	7 (58.3)
Somewhat easy	4 (22.2)	0 (0)	4 (33.3)
Neither easy nor difficult	1 (5.6)	0 (0)	1 (8.3)
Somewhat difficult	1 (5.6)	1 (16.7)	0 (0)
Missing	2 (.)	0 (.)	2 (.)
How relevant was the information in the videos to you as a physician learning about clinical research?, *n* (%)			
Somewhat relevant	12 (66.7)	4 (66.7)	8 (66.7)
Very relevant	6 (33.3)	2 (33.3)	4 (33.3)
Missing	2 (.)	0 (.)	2 (.)
How satisfied were you with the content covered in the videos?, *n* (%)			
Extremely satisfied	6 (33.3)	2 (33.3)	4 (33.3)
Somewhat satisfied	8 (44.4)	3 (50)	5 (41.7)
Neither satisfied nor dissatisfied	3 (16.7)	1 (16.7)	2 (16.7)
Somewhat dissatisfied	1 (5.6)	0 (0)	1 (8.3)
Missing	2 (.)	0 (.)	2 (.)

(Continues)

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TABLE 2 (Continued)

	Overall ($n = 20$)	Non-resident ($n = 6$)	Resident ($n = 14$)
Would you watch more videos developed by our team?, *n* (%)			
No	1 (5.6)	0 (0)	1 (8.3)
Maybe	5 (27.8)	1 (16.7)	4 (33.3)
Yes	12 (66.7)	5 (83.3)	7 (58.3)
Missing	2 (.)	0 (.)	2 (.)
On a scale from 0 to 10, with 0 being would not recommend and 10 being definitely recommend, would you recommend these videos to peers interested in clinical research?, median [IQR]	7 [5.2–8.8]	7.5 [6.2–8.8]	7 [4–8.2]
Overall, how much did these videos increase your knowledge in clinical research?, *n* (%)			
A great deal	2 (15.4)	1 (16.7)	1 (14.3)
A moderate amount	4 (30.8)	2 (33.3)	2 (28.6)
A little	6 (46.2)	3 (50)	3 (42.9)
Not at all	1 (7.7)	0 (0)	1 (14.3)
Missing	7 (.)	0 (.)	7 (.)

Abbreviation: IQR, interquartile range.

than specific mathematical computations. More work is needed to build on these modules and determine how to best integrate this material into residency training programmes.

Ultimately, our main goals are two-fold: (1) to complete resident-led research projects within challenging timeframes and (2) to train residents in research methods and collaboration along the way. In order to be successful, resident collaboration and training must not only be effective but efficient as well. This efficiency is achieved through a highly developed collaboration model and through the delivery of training materials to residents, who tend to be relatively inexperienced in clinical research. Our collaboration model includes multiple templates for residents, which assist them in visualizing what is required and provides them with ready-to-use structure to perform their work and to communicate with biostatisticians on the team. The educational content and delivery is intended to limit time-consuming 1–1 instruction in essential topics and to replace it with asynchronous self-study materials, which were specially designed with this target audience in mind. In essence, what has been described here is an integrated system to help meet the above goals. We continue to develop this collaboration team model and resident training material and have been encouraged by the results to date. We recommend this approach to other biostatistics collaboration units who are engaged in supporting resident-led research.

AUTHOR CONTRIBUTIONS

Clemontina A. Davenport was responsible for the conceptualization; manuscript outline, writing and editing; development and collation of resources; and development, implementation and analysis of survey and evaluation materials. Quentin Ruiz-Esparza and Nicholas Janes were responsible for the production of the video training materials outlined in the manuscript, provided content and critical input on resources and structure and were responsible for manuscript editing. Hui-Jie Lee, Megan Neely, Lacey Rende, Greg Samsa, Kelsey Stilley, Jesse Troy and Tracy Truong provided content and critical input on resources and structure and were responsible for manuscript editing. Steven Grambow and Gina-Maria Pomann were responsible for the initial conceptualization, project design, oversight and workgroup development and manuscript outline, writing and editing.

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CONFLICT OF INTEREST STATEMENT

SCG reports receiving consulting fees from Gilead Sciences for serving on multiple Data Monitoring Committees. JDT reports honoraria from GamidaCell, Synthetic Biologics, Navitas Clinical Research and The EMMES Corporation; consulting fees from AegisCN and The Community Data Roundtable; royalties from SinoCell Technologies and Cryo-Cell International; and research funding from Bristol Myers Squibb and GSK. MLN reports research funding from AstraZeneca and Boehringer Ingelheim. Although these relationships are not perceived to represent a conflict with the present work, they have been included in the spirit of full disclosure. There are no other conflicts of interest to report.

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DATA AVAILABILITY STATEMENT

The authors confirm that the data supporting the findings of this study are available within the article and upon request.

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